Engineering Applications of Artificial Intelligence 162 (2025) 112694

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

# An artificial neural network model for prediction of wall temperature in vertical tube under subcritical and supercritical flow

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ARTICLE INFO

*Keywords:*
Artificial neural network
Subcritical flow
Supercritical flow
Wall temperature prediction
Heat transfer
Dryout

ABSTRACT

Wall temperature prediction in vertical tubes under subcritical and supercritical flow conditions is a challenging task due to involvement of highly non-linear phenomena, such as dryout and heat transfer deterioration (HTD)-induced temperature spikes. Existing literature provides a single framework that addresses each flow condition separately. This study addresses this gap by developing a unified Bayesian Regularization Backpropagation Neural Network (BR-BPNN) model that incorporates critical heat flux conditions with 34 477 and 5535 data points for subcritical and supercritical flows, respectively. The primary goal is to avoid overfitting and achieve optimal generalization across unseen data points while developing the methodology. After a hyperparameter tuning study, the optimized network was found to contain four hidden layers with 125, 100, 75, and 50 neurons. The optimized four-layer network successfully captured the challenging physical behaviors in both flow regimes. It achieved a Root Mean Square Error (RMSE) of 6.57 % and Mean Absolute Error (MAE) of 2.84 % on unseen subcritical testing data, with 96.19 % of predictions falling within a ±10 % error margin. With 97.3 % of predictions falling within the same margin, an RMSE of 4.52 % and MAE of 2.99 % were obtained for supercritical testing data. Important non-linear trends, such as spikes caused by dryout and pseudo-critical behavior, were effectively captured by the model. This model is ideal for incorporation into digital twins, real-time thermal monitoring systems, and system-level thermal-hydraulic codes utilized in nuclear reactors, heat exchangers, and power plant safety analysis.

## 1. Introduction

The application of neural networks in thermal science is emerging as a potential tool for predicting complex processes such as Critical Heat Flux (CHF) and heat transfer under various fluid flow conditions. CHF prediction has always been a topic of active research because of its importance in avoiding catastrophic events, such as fuel rod overheating and meltdown, in applications such as nuclear reactors, heat exchangers, and power plants. The artificial neural network (ANN) is a promising alternative to predict heat transfer compared to conventional methods like Computational Fluid Dynamics (CFD) and mechanistic models. ANNs can identify patterns that could be missed by traditional approaches, owing to their accuracy and wider applicability, as large datasets are available for training.

Dryout is the most common cause of CHF during subcritical vertical tube flow boiling (<22.1 MPa). The dryout phenomenon creates an annular flow pattern in which a thin layer of liquid adheres to the wall, while vapor flows through the central region. As evaporation continues along the heated length, the liquid film becomes progressively thinner until it can no longer maintain wall coverage, ultimately breaking apart owing to evaporation or instability. This results in the complete loss of the liquid film, as illustrated in Fig. 1(a), exposing the surface directly to vapor with low thermal conductivity. The inability to rewet the surface leads to a continuous and steep increase in the wall surface temperature, which triggers dryout. This process indicates a more severe CHF than that encountered in departure from nucleate boiling (DNB), resulting in a permanent degradation of heat transfer efficiency.

Under supercritical pressure conditions (above 22.1 MPa), water exhibits single-phase fluid behavior without a distinct phase change; however, it undergoes significant alterations in its thermophysical properties near the pseudo-critical temperature. As the bulk fluid approaches this pseudo-critical temperature, the specific heat capacity reaches its maximum and turbulence intensifies, facilitating transient increases in heat transfer. However, if the wall temperature exceeds this pseudo-critical point, a low-density region forms close to the wall, as shown in Fig. 1(b), which inhibits turbulent mixing and convective heat

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https://doi.org/10.1016/j.engappai.2025.112694
Received 8 April 2025; Received in revised form 26 September 2025; Accepted 6 October 2025
Available online 13 October 2025
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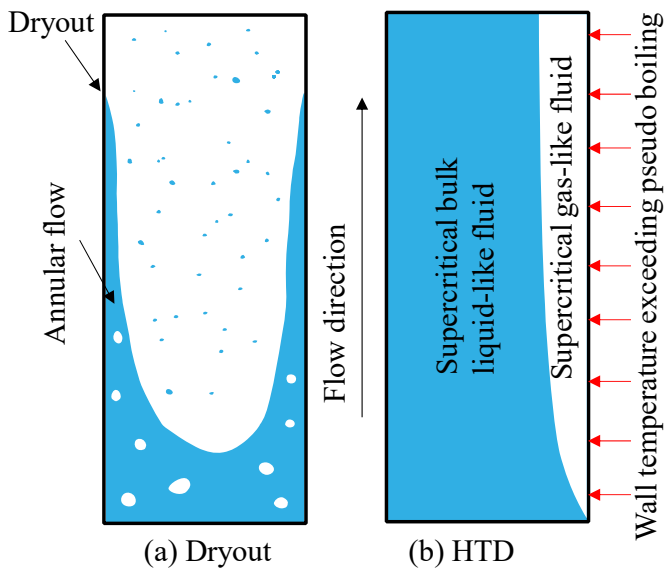


**Fig. 1.** Illustrative representation of critical heat transfer phenomenon: (a) dryout observed in subcritical flow and (b) heat transfer deterioration (HTD) due to pseudo-boiling observed in supercritical flow.

transfer. This phenomenon results in thermal stratification, where continued heating leads to a marked reduction in the heat transfer coefficient, causing a sudden and sustained increase in wall temperature. This phenomenon, known as Heat Transfer Deterioration (HTD), resembles the thermal behavior observed during dryout but arises from property-driven mechanisms in single-phase supercritical flow, rather than from film drying.

Liang et al. (2021) presented a deep ANN model to predict boiling heat transfer in helical coils under high-gravity conditions. They used 531 experimental data points for training and considered input variables such as mass flow rate, thermal power, inlet temperature, and helical coil diameter. The ANN had three hidden layers and achieved an $R^2$ value of 0.853. Kim et al. (2021) presented a machine learning model to predict CHF in narrow rectangular channels under steady-state conditions by combining the Deep Belief Network (DBN) and Residual Network (ResNet). The pseudo-data generated using CHF correlations were used for training, and the model achieved a mean square error of 17.2 % for the test data and 26.4 % for the experimental data not included in the training. Their model effectively predicted CHF value but did not predict the spatial location where CHF occurred. Li et al. (2014) utilized the backpropagation neural network (BPNN) method to conduct a sensitivity analysis of CHF under flow instability. They reported that oscillation amplitude had the highest impact on CHF occurrence, whereas the inlet temperature had the smallest impact. Their BPNN model exhibited errors within 5 % of experimental data. Park et al. (2020) developed an ANN for the prediction of wall temperature in subcritical flow conditions with root mean square error (RMSE) within ±2K. They reduced the computational time by 86 % by implementing these trained models in the Safety and Performance Analysis Code for Nuclear Power Plants (SPACE) code. Their model was trained on data generated from simulation results rather than directly sourced from experimental reports. Chang et al. (2018) utilized 5380 data points gathered from a well-recognized lookup table (Loewenberg et al., 2008) to train an ANN under supercritical flow conditions. Their model exhibited excellent accuracy in wall temperature predictions and had good general applicability. Lin et al. (2022) utilized 7349 experimental data points of the flow boiling heat transfer coefficient for 16 types of refrigerants to develop a novel neural network architecture. Their model showed a mean absolute error (MAE) of 13.8 % and good accuracy for diverse testing conditions.

Gouravaraju et al. (2023) presented a Bayesian regularization-based backpropagation neural network (BR-BPNN) to predict peeling behaviors such as the maximum normal and tangential pull-off forces, resultant force angle, and displacements at detachment for gecko spatula peeling. They used Finite Element (FE) data for training. They used k-fold cross-validation during training to evaluate the model performance. Their BR-BPNN model achieved average relative errors of 0.24 % for forces and 1.22 % for displacements. Mudawar et al. (2024a) presented an ANN to predict CHF phenomena in rectangular channels using an experimental database collected from the International Space Station (ISS) and Earth-based experiments. Their trained model achieved an MAE of 7.99 % for predicting the Nusselt number. 96 % of their dataset was predicted within an error margin of ±30 %, while 99 % of the dataset was predicted within an error margin of ±50 %. Their study utilized a deeper neural network with 13 hidden layers, which could exhibit overfitting, and they did not use techniques like cross-validation to minimize the risk of overfitting. Qiu et al. (2024) developed three types of data-driven models for CHF prediction using the Back Propagation Neural Network (BPNN), Random Forest (RF), and Physics-Informed Machine Learning (PIML) methodologies. The BPNN model achieved the highest accuracy with an average relative error of 3.64 %, outperforming traditional CHF prediction models. The optimized BPNN model was integrated into the Advanced Reactor Safety Analysis Code (ARSAC) system and validated against experimental data from Oak Ridge National Laboratory-Thermal Hydraulic Test Facility (ORNL-THTF) experimental data for the wider application of nuclear reactors. Sajjad et al. (2024) developed a physics- and correlation-informed deep learning model for the prediction of pool boiling regimes, such as nucleation boiling, transition boiling, and CHF, with an $R^2$ of 0.98. They used lower-order boiling data, surface morphology, liquid properties, and different operating conditions to train a single model that exhibited good accuracy during testing across diverse operating conditions. However, their model was trained and tested on data from a single experimental setup, which could have led to bias in the trained model. Aliyu et al. (2023) utilized 1367 data points for annular gas-liquid two-phase flows to predict the entrained droplet fraction in vertical pipes. They used different input parameters, such as superficial velocities, fluid properties, and pipe diameter, to train the model and achieved an $R^2$ of 0.98. They used blind data to validate their model and concluded that superficial velocity has a greater impact on droplet entrainment fraction. Niu et al. (2024) coupled the Deep Feed-Forward Neural Network (DFNN) with CHF) empirical correlations to predict both the value and location of CHF in narrow rectangular channels. With the addition of empirical correlations into the neural network, the developed model enhanced the prediction accuracy by reducing error from 10 % to 5 % compared to the traditional neural network method. Their model was trained on CFD simulation data due to experimental measurement difficulties, which could prohibit the deployment of the model in the real world. Wang et al. (2025) compiled 660 experimental CHF data points for narrow rectangular channels, showing that existing empirical correlations perform poorly, and instead trained four machine-learning models. Their backpropagation model significantly outperformed other models. Their model predicted only the CHF as a point value and not the spatial variation in the wall temperature or local boiling behavior.

Traditional thermal hydraulics for CHF prediction has advantages and disadvantages. Phenomenological models rely on simplified assumptions and often fail under extreme flow conditions. Empirical correlations are frequently confined to narrow operating conditions and often struggle to capture the highly non-linear behaviors observed during dryout and heat transfer deterioration. CFD is a versatile tool that can be applied to a wide range of operating conditions and geometrical variations. However, it is computationally expensive, particularly for real-time monitoring aimed at preventing hazardous conditions. This is precisely where Deep Learning (DL) methods are used. DL methods utilize comprehensive datasets with multivariate mapping from

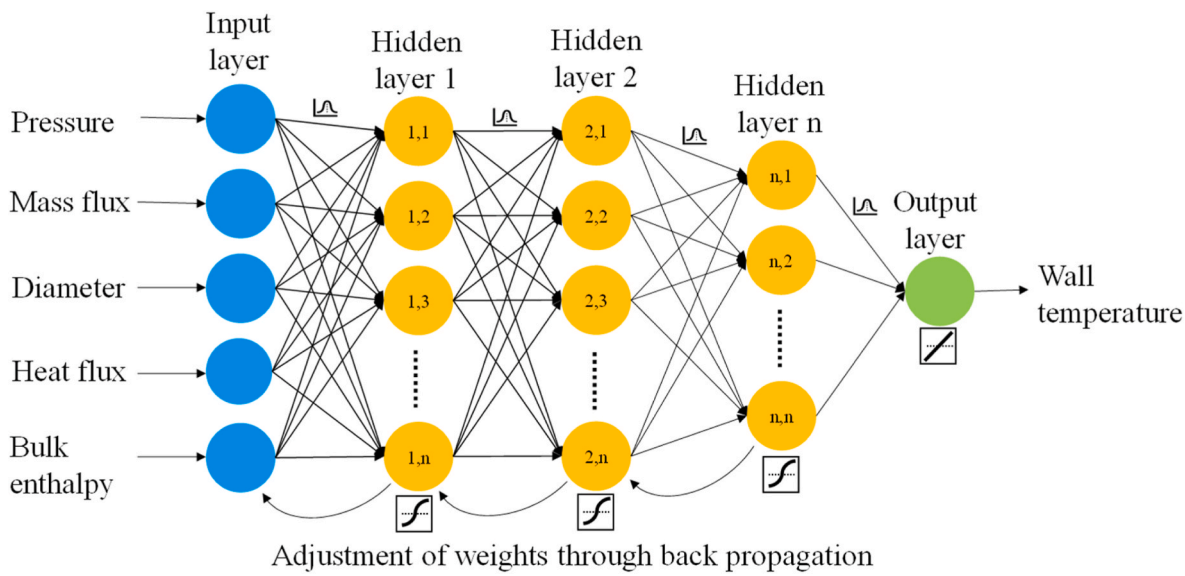


**Fig. 2.** Proposed BR-BPNN architecture showing input and output parameters along with the activation functions for the hidden layers and output layer.

operational parameters to predict wall temperature. Moreover, a DL model can easily accommodate the combined data from subcritical and supercritical CHF regimes, providing accurate predictions of temperature spikes, with results obtained within milliseconds.

Although ANNs have demonstrated their effectiveness in predicting CHF under subcritical conditions, the current literature reveals two significant research gaps that require further development. First, existing studies have primarily focused on CHF value prediction, but only limited attention has been paid to the wall temperature profiles, particularly in the prediction of sharp temperature jumps at dryout locations for subcritical flow conditions. Second, no study to date has developed a single unified model capable of handling both subcritical and supercritical flow conditions for a diverse range of input conditions, such as pressure, mass flow rate, tube diameter, heat flux, and bulk enthalpy.

A significant difference is that an artificial neural network (ANN) is based on data, while traditional mechanistic, physics-based models solve the governing conservation equations directly. Computational fluid dynamics (CFD) and physics-informed neural networks (PINNs) simulate fluid behavior by enforcing these equations. An artificial neural network (ANN), on the other hand, learns the highly nonlinear mapping from operating inputs to wall temperature directly from large experimental datasets, without explicitly resolving the underlying physical processes. In this framework, the physical mechanisms are not predetermined; rather, they are implicitly encoded within the data and represented by the learned network parameters.

To bridge this research gap, the present study developed a single unified BR-BPNN model to train on combined data for both subcritical and supercritical flows, focusing on the prediction of phenomena of interest, such as dryout and Heat Transfer Deterioration (HTD), using large datasets over a diverse range of operating conditions gathered from different experimental studies. This model can predict the wall temperature within a fraction of a second and can be utilized for real-time thermal system monitoring to prevent accidents.

Section II details the development of the proposed BR-BPNN, focusing on the network architecture, normalization methods, training procedure, and hyperparameter tuning process. It provides methodological details on the implementation of k-fold cross-validation and the datasets used for training and testing. Section III presents the results and discussion, including the performance evaluation of the model for both subcritical and supercritical conditions and its comparison with experimental data. Finally, Section IV summarizes the main conclusions of this study and discusses real-world implications and future research.

## 2. Artificial neural network model development

This section describes the methodology and implementation of the BR-BPNN, including its architecture, mathematical framework, and data preprocessing techniques, such as normalization and feature selection. In addition, the k-fold cross-validation strategy and the process of hyperparameter tuning are explained in detail to determine the best-performing neural network architecture. The proposed BR-BPNN model was developed using MATLAB (Beale et al., 2020).

### 2.1. Bayesian Regularization Backpropagation Neural Network

The BR-BPNN (MacKay, 1992) can provide better predictions than traditional neural networks because it utilizes Bayesian statistical principles. Bayesian regularization provides a better generalization capability and reduces the occurrence of overfitting. The BR-BPNN strikes a balance between neural network model complexity and prediction accuracy with the help of a probabilistic approach (Demuth et al., 2014). The backpropagation algorithm shown in Fig. 2 trains the network to learn from the input features to the output targets by iteratively updating the weights and biases to minimize the sum of squared errors between the forecasted and actual target outputs. Backpropagation updates the weights and biases based on the gradients of the performance metrics determined from the predicted and actual values. In addition, Bayesian regularization enhances conventional backpropagation by integrating a probabilistic approach for the regularization of the network's weights. This is achieved by adding a penalty term to the performance metrics, which controls the complexity of the model. Using this approach, Bayesian regularization avoids overfitting and ensures that the trained network maintains better generalization performance for unseen data (Gouravaraju et al., 2023; MacKay, 1992; Burden and Winkler, 2008). A detailed mathematical explanation of the BR-BPNN approach is provided below.

Consider a training dataset, where $x_i$ are the inputs and $y_i$ are the target outputs. In the present study, the inputs are pressure ($P$), mass flow rate ($G$), diameter ($D$), heat flux ($Q$), and bulk enthalpy ($H$). The target output is wall temperature ($T_w$). The sum of the squared error ($E_D$) is defined as

$$E_D = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{1}$$

where $\hat{y}_i$ denotes the predicted output value. Let $W$ represent the vector of all weights and biases in the network. The weight penalty is the sum of the squared network weights ($E_W$) and can be expressed as

$$E_W = \sum_{i=1}^{n} (W)^2 \tag{2}$$

The overall objective function $F(W)$ becomes

$$F(W) = \alpha E_W + \beta E_D \tag{3}$$

where $\alpha$ and $\beta$ are regularization parameters controlling the trade-off between minimizing weights and fitting the data. A large $\alpha$ emphasizes the weight-decay term and reduces over-fitting, whereas large $\beta$ risks over-fitting, and vice versa. In Bayesian regularization, $\alpha$ and $\beta$ are treated as random variables whose values are re-estimated at every training epoch so that the posterior probability of the weights is maximized (MacKay, 1992). They are not set a-priori by the user.

Bayesian regularization treats the weights $W$ as random variables and applies Bayes' theorem to maximize the posterior distribution for the given data $D$.

$$P(W|D,\alpha,\beta,M) = \frac{P(D|W,\beta,M)\, P(W|\alpha,M)}{P(D|\alpha,\beta,M)} \tag{4}$$

where $P(D|W,\beta,M)$ is a likelihood function considering that noise in the training dataset follows a Gaussian distribution and can be written as

$$P(D|W,\beta,M) = \frac{e^{(-\beta E_D)}}{Z_D(\beta)} \tag{5}$$

where $P(W|\alpha,M)$ is the prior probability density function for network weights with a Gaussian distribution and can be written as

$$P(W|\alpha,M) = \frac{e^{(-\alpha E_W)}}{Z_W(\alpha)} \tag{6}$$

In Equations (5) and (6), $Z_D(\beta) = \left(\frac{\pi}{\beta}\right)^{n/2}$ and $Z_W(\alpha) = \left(\frac{\pi}{\alpha}\right)^{N/2}$. $W$ is a vector of size $N$ containing all the weights and biases of the network. $D$ is a training dataset with $n$ inputs and target outputs. $M$ is the particular neural network model. $P(D|\alpha,\beta,M)$ is the normalization factor in Bayesian inference for the posterior probability distribution of the weights $W$ in a neural network. It ensures that the posterior probability integrates to 1. $Z_F(\alpha,\beta)$ is the normalization constant.

$$P(D|\alpha,\beta,M) = \frac{Z_F(\alpha,\beta)}{Z_D(\beta) Z_W(\alpha)} \tag{7}$$

Substituting Equation (6) into the posterior distribution (Equation (4)), we get

$$P(W|D,\alpha,\beta,M) = \frac{e^{-F(W)}}{Z_F(\alpha,\beta)} \tag{8}$$

The optimal weights maximize the posterior probability function by using the $e^{-F(W)}$ term. $F(W)$ combines the regularization term ($E_W$) and data error term ($E_D$). Maximizing the posterior is equivalent to minimizing $F(W)$.

The complexity of the model $M$ is determined by the regularization parameters $\alpha$ and $\beta$, which must be estimated from the data. To achieve this, Bayes' rule was applied once more to optimize these parameters:

$$P(\alpha,\beta|D,M) = \frac{P(D|\alpha,\beta,M)\, P(\alpha,\beta|M)}{P(D|M)} \tag{9}$$

The term $P(\alpha,\beta|D,M)$ represents the assumption of a uniform posterior distribution for the parameters $\alpha$ and $\beta$. From the above equation, it becomes clear that maximizing the likelihood function $P(D|\alpha,\beta,M)$ also maximizes the posterior probability $P(\alpha,\beta|D,M)$. Additionally, it is important to note that the likelihood function in Equation (9) serves as the normalization factor for Equation (4). To evaluate the unknown normalization constant $Z_F(\alpha,\beta)$ in Equation (8), we expand the objective function $F(W)$ using a second-order Taylor approximation around the minimum point $W^*$, where the gradient is zero. This yields

$$F(W) = F(W^*) + \frac{1}{2}(W - W^*)^T H(W^*)(W - W^*) \tag{10}$$

Here, $H(W^*) = \nabla^2 F(W^*)$ is the Hessian matrix of second derivatives evaluated at $W^*$. In the case of Bayesian regularization, this becomes

$$H = \beta \nabla^2 E_D + \alpha \nabla^2 E_W \tag{11}$$

Because $H(W^*)$ is symmetric and positive definite, it can be decomposed into $H = \left(H^{1/2}\right)^T H^{1/2}$. We now make a change of variables to simplify the Gaussian integral:

$$u = H^{1/2}(W - W^*) \tag{12}$$

Substituting Equations (10) and (12) into Equation (8) and integrating both sides, we obtain

$$1 = \frac{1}{Z_F(\alpha,\beta)} e^{-F(W^*)} \int_{\mathbb{R}^N} e^{-u^T u/2} \left[ |H(W^*)|^{-1} \right]^{1/2} \tag{13}$$

Solving Equation (13), we get

$$Z_F(\alpha,\beta) = 2\pi^{N/2} |H|^{-1/2} e^{-F(W^*)} \tag{14}$$

Substituting the above equation of $Z_F(\alpha,\beta)$ and the previously defined $Z_D(\beta)$ and $Z_W(\alpha)$ into the marginal likelihood function (Equation (7)), we get

$$P(D|\alpha,\beta,M) = \left(\frac{\beta}{2\pi}\right)^{n/2} \left(\frac{\alpha}{2\pi}\right)^{N/2} |H|^{-1/2} e^{-F(W^*)} \tag{15}$$

By taking the log on both sides, the log-marginal likelihood equation can be written as

$$log(P(D|\alpha,\beta,M)) = \frac{n}{2} log\, \beta + \frac{N}{2} log\, \alpha - F(W^*) - \frac{1}{2} log|H| \tag{16}$$

To find optimal values of $\alpha$ and $\beta$, we can equate the derivative of nature logarithm of the above equation to zero to get

$$\frac{\partial logP}{\partial \alpha} = 0 \Longrightarrow \alpha^* = \frac{\gamma}{2E_w(W^*)} \tag{17}$$

$$\frac{\partial logP}{\partial \beta} = 0 \Longrightarrow \beta^* = \frac{n - \gamma}{2E_D(W^*)} \tag{18}$$

where $\gamma = N - \alpha^*\, trace^{-1}(H^*)$, which denotes the number of effective parameters

and quantifies how many network parameters are actively involved in reducing the error function. $H^*$ represents the Hessian matrix of the objective function evaluated at $W^*$, which is computed using the Gauss-Newton approximation as follows:

$$H^* \approx J^T J \tag{19}$$

where $J$ is the Jacobian matrix after determining the first derivative of the network errors with respect to the network weights.

At the end of training, it is essential to perform certain checks on the number of effective parameters to ensure optimal network performance. The computation of the Hessian matrix ($H^*$) at the minimum point $w$ is inherently addressed by the Levenberg-Marquardt (LM) training algorithm during the minimization of $F(W)$. In the Levenberg-Marquardt algorithm, the network weights and biases are updated at the $\kappa^{th}$ iteration using the following approach:

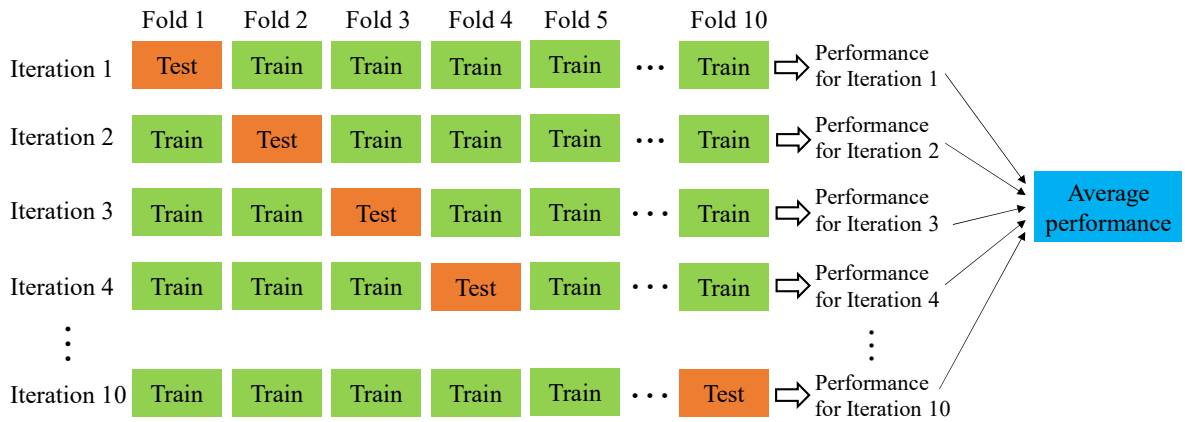


**Fig. 3.** K-fold cross validation methodology.

$$w^{\kappa+1} = w^{\kappa} - \left[J^T J + \lambda I\right]^{-1} J^T e \tag{20}$$

where $\lambda$ represents the Levenberg's damping factor and $J^T e$ is the gradient of the errors, which is expected to be near zero at the end of training.

### 2.2. K-fold cross-validation methodology

The k-fold cross-validation approach was implemented using 10 folds to rigorously assess the model performance and ensure generalizability, as shown in Fig. 3. The dataset was randomly shuffled and divided into ten equal subsets, with nine folds used for training and one for testing in each iteration. This process was repeated systematically with each fold serving as the test set exactly once. This method ensures that every data point contributes to both the training and validation phases. For each iteration, a fresh neural network was initialized and trained for a maximum of 25 000 epochs. To prevent overfitting and reduce training time, an early stopping mechanism was incorporated. The training process was terminated if no performance improvement was observed over eight consecutive epochs.

Both the input and output data were normalized to the range [–1, 1] before the training started. This normalization step helps improve the numerical stability of the model training, and the network can learn more effectively by avoiding potential issues related to varying magnitudes between different input values. Each hidden layer uses a hyperbolic tangent activation ($f(x) = \frac{2}{1+e^{-2x}} - 1$), which introduces nonlinearity and maps data between −1 and 1. Finally, the output layer uses a linear activation function ($f(x) = x$), which ensures that the network produces continuous values suitable for regression tasks like predicting the target output values of wall temperatures. The neural network training and testing were performed on a high-performance computing cluster comprising three compute nodes. Each node was equipped with dual Intel Xeon E5-2697 v4 CPUs running at 2.30 GHz, providing a total of 36 physical cores per node. No GPU acceleration was used during the training. Each node was equipped with 256 GB of RAM.

The cross-validation loop continues until each of the 10 folds is used once as the validation set, which results in 10 different trained models, and their corresponding performance metrics are saved. Model performance was evaluated using RMSE, Mean Absolute Error (MAE), and coefficient of determination ($R^2$). The performance metrics collected at the end of each iteration are averaged to assess the overall accuracy of the trained model. This method ensures that the performance of the model is not influenced by a specific data split. Each metric can be mathematically expressed as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{21}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{22}$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \tag{23}$$

where $y_i$ is the actual target output value, $\hat{y}_i$ is the predicted target output value, and $n$ is the number of observations. $SS_{res}$ (sum of squared residuals) $= \sum_1^n (y_i - \hat{y}_i)^2$ and $SS_{tot}$ (total sum of squares) $= \sum_1^n (y_i - \bar{y}_i)^2$, where $\bar{y}_i$ is the mean of $y_i$.

### 2.3. Experimental database gathered for training and testing

Table 1 summarizes the large database used for training and testing,

**Table 1**
Database used for training and testing.

| Source | $P$ (bar) | $G$ (kg/m$^2$s) | $D$ (mm) | $q$ (W/cm$^2$) | Number of data points | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Training | | Testing | |
| | | | | | Subcritical | Supercritical | Subcritical | Supercritical |
| (Becker et al., 1983) | 29.8–201 | 496–3113 | 10–24.7 | 9–129.5 | 24851 | – | 52 | – |
| (Herkenrath et al., 1967) | 140–215 | 700–3500 | 10–20 | 20–180 | 9027 | – | 53 | – |
| (Loewenberg et al., 2008) | 225–250 | 700–3500 | 8–20 | 30–160 | – | 5265 | – | 14 |
| (Bennett et al., 1967) | 69 | 390–1356 | 12.6 | 46–82 | – | – | 97 | – |
| (Wang et al., 2011) | 154–280 | 600–1200 | 19.8–26 | 35–66 | 39 | – | 35 | 45 |
| (Shen et al., 2014) | 190–280 | 800–1536 | 17 | 25–46.8 | – | 20 | 58 | – |
| (Xu, 2004) | 230–250 | 1200 | 12 | 60 | – | – | – | 51 |
| (Zhu et al., 2009) | 130–300 | 400–1200 | 26 | 20–40 | 261 | 190 | – | – |
| (Shen et al., 2016) | 110–320 | 170–800 | 19 | 8.5–42 | 299 | 60 | – | – |
| (Lei et al., 2017) | 280 | 600 | 26 | 30 | – | – | – | 35 |

**Table 2**
Effects of combinations of training algorithms and input layer activation functions on errors.

| Training algorithm | Input activation function | Normalization Technique | RMSE (°C) | MAE (°C) | R-square |
|---|---|---|---|---|---|
| *trainlm* | *poslin* | *mapminmax* | 30.68 | 19.46 | 0.8721 |
| | *radbas* | *[-1,1]* | 29.44 | 19.14 | 0.8819 |
| | *logsig* | | 26.20 | 16.29 | 0.9066 |
| | *tansig* | | 25.96 | 16.01 | 0.9253 |
| | | *Mapstd [0,1]* | 26.47 | 16.24 | 0.9048 |
| *trainscg* | | *mapminmax* | 44.81 | 30.79 | 0.7264 |
| *trainbr* | | *[-1,1]* | 25.05 | 15.21 | 0.9144 |

**Table 3**
Performance of testing dataset averaged at the end of 10-fold cross-validation for various neural network architectures.

| Model No. | Layers and neurons | RMSE (°C) | MAE (°C) | R-Square |
|---|---|---|---|---|
| 1 | [10] | 41.95 | 27.39 | 0.7611 |
| 2 | [50] | 28.53 | 17.56 | 0.8893 |
| 3 | [100] | 25.05 | 15.21 | 0.9144 |
| 4 | [75 50] | 13.23 | 6.13 | 0.9759 |
| 5 | [100 75 50] | 11.67 | 4.53 | 0.9811 |
| 6 | [125 100 75 50] | 11.48 | 4.16 | 0.9818 |
| 7 | [200 100 50] | 12.44 | 4.79 | 0.9788 |
| 8 | [150 125 100 75] | 12.07 | 4.31 | 0.9798 |
| 9 | [200 150 100 50] | 12.22 | 4.53 | 0.9795 |

sourced from various experimental studies on subcritical and supercritical flows. All training data from the subcritical and supercritical regimes were combined and supplied as a single input dataset during neural network training. The training process was performed using only the training data, and the testing data were blinded to ensure that the robustness of the generalization for the developed model under unseen conditions could be strictly judged. Some of the testing data were collected from independent and separate research (Bennett et al., 1967; Xu, 2004; Lei et al., 2017), which were put aside exclusively for testing purposes with no data points used for training the neural network model. The training dataset contained 34 477 data points for the subcritical condition and 5535 data points for the supercritical condition. There were 295 data points for the subcritical condition and 145 data points for the supercritical condition, considered as blind (unseen) data for isolated testing.

### 2.4. Hyperparameter tuning

Before proceeding to hyperparameter tuning of the network architecture, a preliminary study was conducted to identify the optimal combination of activation functions and training algorithms. This initial analysis aimed to ensure that the subsequent ANN model has a solid foundation. This study utilized a simple network with a single hidden layer containing 100 neurons, and the results are summarized in Table 2. In the first phase, various activation functions like Rectified Linear Unit (*poslin*), Radial Basis Transfer Function (*radbas*), Log-sigmoid (*logsig*), and hyperbolic tangent sigmoid (*tansig*) were tested using the Levenberg–Marquardt (*trainlm)* algorithm with data normalization between −1 and 1. Among these, *tansig* demonstrated the best performance, achieving an RMSE) of 25.96 °C and $R^2$ value 0.9253. Further, another popular normalization method of Z-score standardization (*mapstd*) was used, and the results clearly demonstrated that the model performance was degraded as the RMSE increased from 25.82 °C to 26.47 °C. With *tansig* selected as the preferred activation function and normalization within [−1,1], various other training algorithms were evaluated: Levenberg–Marquardt (*trainlm*), Scaled Conjugate Gradient (*trainscg*), and Bayesian Regularization (*trainbr*). The *trainbr* algorithm outperformed the other training methods, achieving the lowest RMSE of 25.05 °C and highest $R^2$ value of 0.9144, making it the most suitable choice for developing the final BR-BPNN model. A key advantage of *trainbr* is its ability to not only perform backpropagation for training but also automatically adjust the regularization parameters by minimizing a combined objective function that accounts for both the squared error and magnitude of the network weights. This eliminates the need for manual tuning of the regularization hyperparameters. Internally, it uses a modified Levenberg–Marquardt optimization scheme and requires no external optimizer tuning because it effectively balances model complexity and prediction accuracy through Bayesian inference. This makes it particularly efficient and reliable for training on complex datasets without the need for extensive optimizer-related hyperparameter tuning.

Hyperparameter tuning for the network architecture was conducted

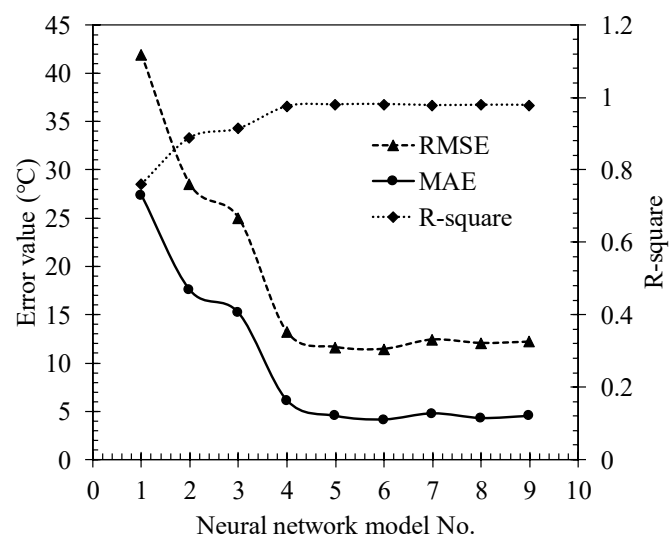


**Fig. 4.** Performance of testing dataset averaged at the end of k-fold cross-validation for various ANN models.

by varying the layer count (network depth) and neurons per layer (network width) using the performance metrics averaged and collected at the end of the 10-fold cross-validation method with the BR-BPNN model, as shown in Table 3 and Fig. 4. The RMSE, MAE, and $R^2$ values were used as primary performance metrics.

The initial trials were conducted using a Single-Layer Network (SLN) with different numbers of neurons for Models 1–3. Increasing the number of neurons in a single layer resulted in a substantial improvement in the performance metrics, as shown in Table 3 and Fig. 4. Model 1, consisting of 10 neurons, showed the lowest performance with an RMSE of 41.95 °C, an MAE of 27.39 °C, and $R^2$ of 0.7611. This indicates that Model 1 does not possess sufficient neurons to accurately learn the given dataset patterns. However, increasing the neuron count to 50 (Model 2) and 100 (Model 3) led to significant improvements in the performance metrics. Model 3 resulted in an RMSE of 25.05 °C, which is relatively lower than the RMSE of Model 2 (28.53 °C). Similarly, Model 3 showed an MAE of 15.21 °C, relatively smaller than the MAE of Model 2 (17.56 °C). In addition, Model 3 showed a significant improvement in $R^2$, from 0.8893 to 0.9144, compared to Model 2. Further increasing the number of neurons in a single-layer network did not significantly improve the performance metrics, indicating a possible limitation of the learning ability of the (SLN).

A multi-layer network (MLN) (Models 4–9) was examined to see an improvement in predictive accuracy, as shown in Table 2. Adding more hidden layers resulted in a significant performance enhancement. Model 4 with a two-layer network [75 50] was found to significantly outperform the best SLN (i.e., Model 3). Model 4 showed an RMSE of 13.23 °C, MAE of 6.13 °C, and $R^2$ of 0.9759. This can be explained by the fact that

**Table 4**
Architecture of the optimized BR-BPNN Model 6.

| Layer | Type | Number of Neurons | Activation Function |
|---|---|---|---|
| 1 | Input | 5 | – |
| 2 | Hidden Layer 1 | 125 | Hyperbolic Tangent Sigmoid (*tansig*) |
| 3 | Hidden Layer 2 | 100 | Hyperbolic Tangent Sigmoid (*tansig*) |
| 4 | Hidden Layer 3 | 75 | Hyperbolic Tangent Sigmoid (*tansig*) |
| 5 | Hidden Layer 4 | 50 | Hyperbolic Tangent Sigmoid (*tansig*) |
| 6 | Output | 1 | Linear (*purelin*) |

multiple layers provide hierarchical feature extraction, which allows the model to acquire more abstract and complex representations of the input data. Increasing the multilayers to three layers with Model 5 [100 75 50], the performance was found to improve moderately, with an RMSE of 11.67 °C, MAE of 4.53 °C, and $R^2$ of 0.9811. A further increase in multilayers with Model 6 [125 100 75 50] showed a slight improvement in performance, with an RMSE of 11.48 °C, MAE of 4.16 °C, and $R^2$ of .9818.

Further increasing the network depth and width beyond Model 6 resulted in no significant improvements from Models 7 to 9, with a slight performance degradation in some cases, as shown in Fig. 4. This model can improve performance up to a certain value, and then performance drops owing to overfitting or the inability to learn meaningful patterns. This may cause the model to capture noise and substantially increase training time, which is disadvantageous. Therefore, Model 6 [125 100 75 50] was chosen as the final model for wall temperature predictions and further validations. The hyperparameter tuning study of the neural network width and depth shows that it is important not only to identify the best model based on performance metrics but also to avoid the utilization of too many layers because it can lead to less meaningful results. The detailed layer-by-layer structure of this optimal model is summarized in Table 4.

After completing the cross-validation, reporting the performance metrics, and choosing the best architecture (i.e., Model 6) as shown in

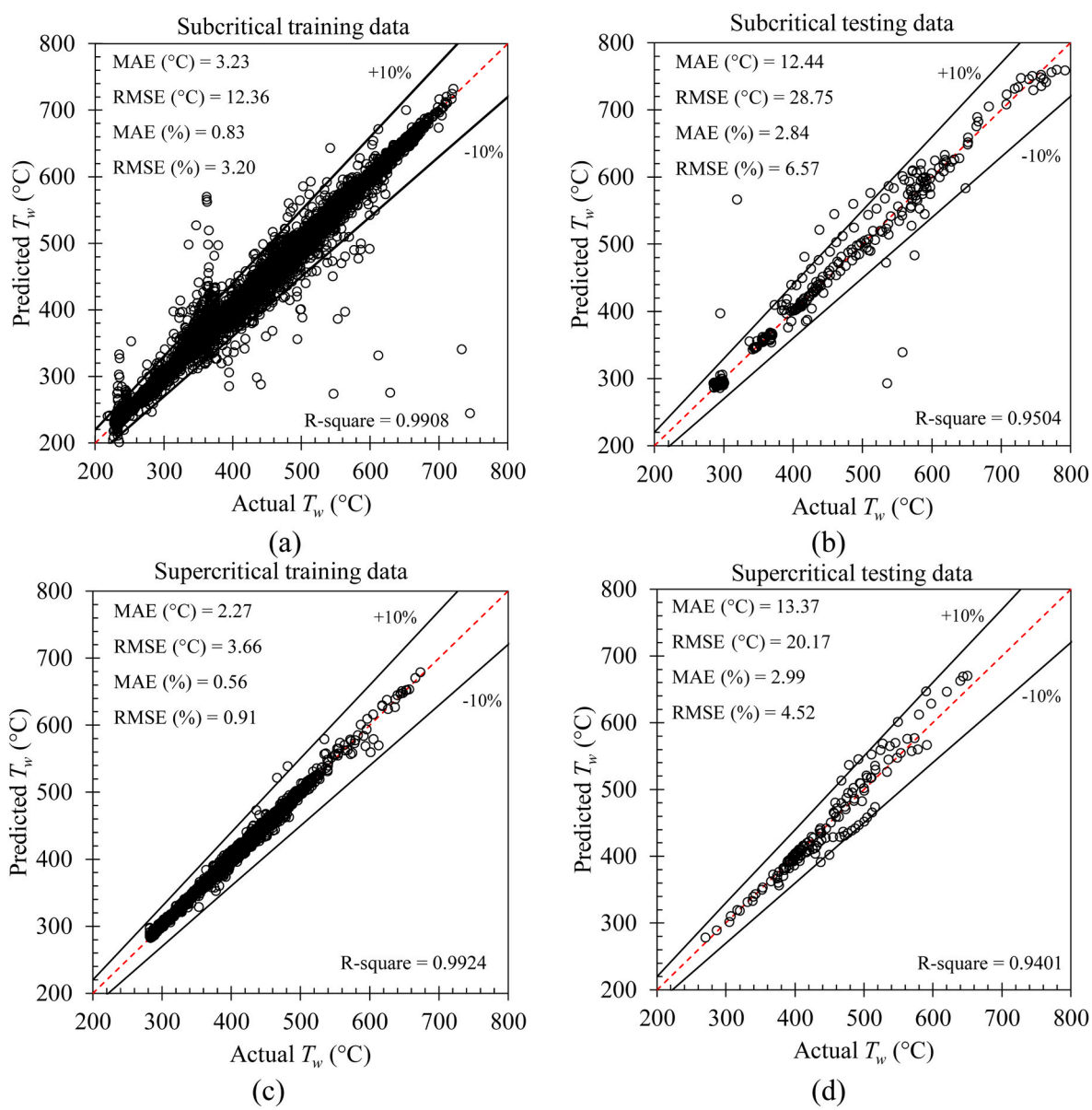


**Fig. 5.** Scatter plots of actual vs predicted $T_w$ (°C) for (a) subcritical training data points, (b) subcritical testing data points, (c) supercritical training data points, and (d) supercritical testing data points.

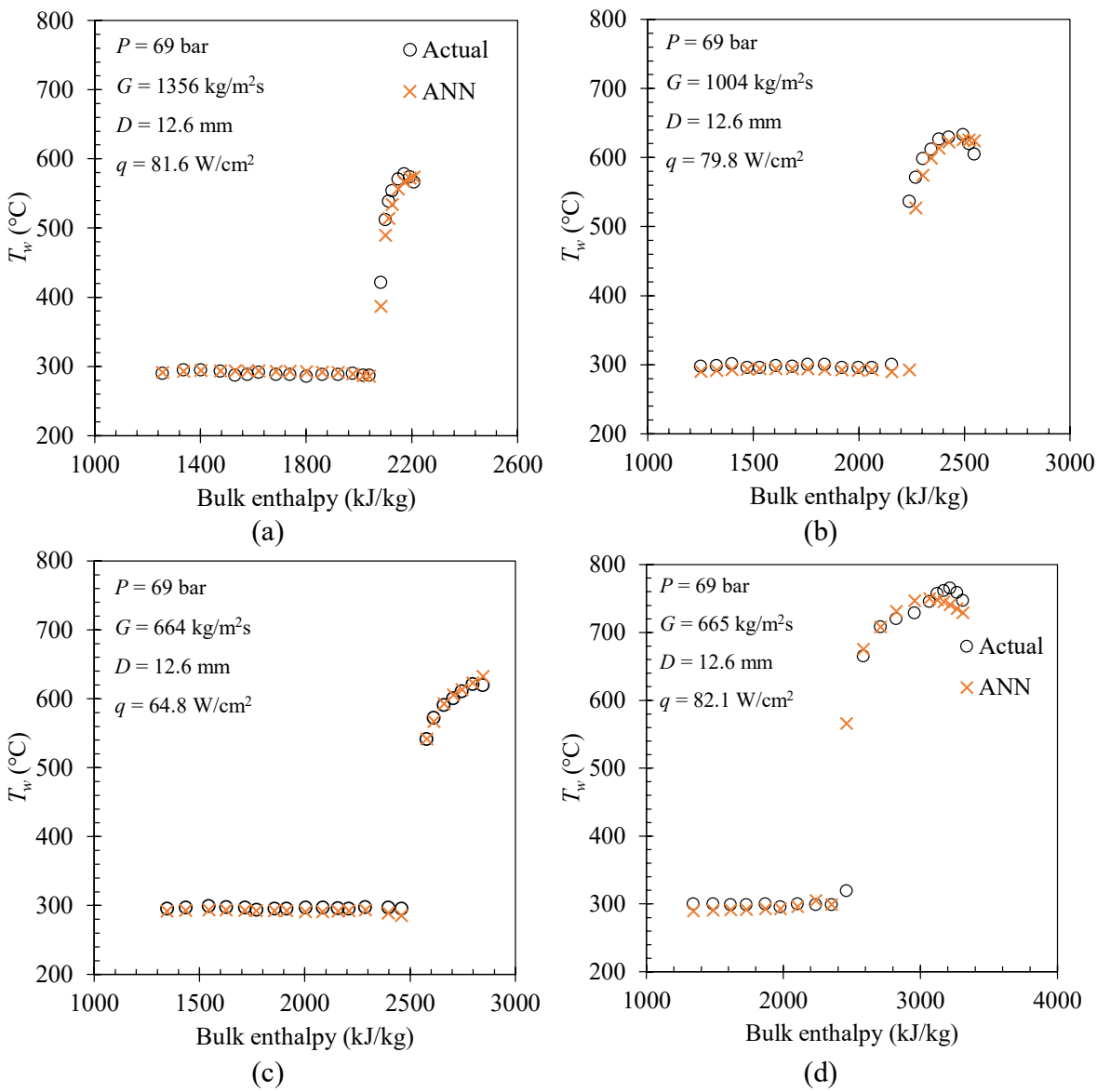


**Fig. 6.** Comparison of actual vs BR-BPNN-predicted wall temperature over given bulk enthalpy for subcritical water based on experimental data of (Bennett et al., 1967).

Table 4, the network was retrained on the entire training dataset to maximize the amount of data available for training. The method of training on a full dataset after cross-validation and hyperparameter tuning is commonly followed as standard practice in machine learning applications (Géron, 2019; Varma and Simon, 2006). The final model took approximately 20 h to train on stated computational machine of three nodes. This final model benefits from the insights gained during the cross-validation step and was optimized for general use. The activation functions remain consistent, with hyperbolic tangent sigmoid functions applied to the hidden layers and a linear function used in the output layer during final training. The final trained model is then saved for future predictions, and the performance metrics are recorded to provide a benchmark of the model's accuracy, as discussed in Section III.

## 3. Results and discussion

This section reports the performance of the developed BR-BPNN model in predicting wall temperatures under subcritical and supercritical flow conditions. The accuracy of the predictions was analyzed based on experimental datasets from multiple sources and key performance metrics such as RMSE, MAE, and $R^2$.

### 3.1. Scatter plot assessing prediction capability over training and testing data

Fig. 5(a–d) compare the actual experimental data (Table 1) with the predicted data from the developed BR-BPNN. The scatter plots are divided into four segments: (a) subcritical training data, (b) subcritical testing data, (c) supercritical training data, and (d) supercritical testing data. The solid black lines mark a ±10 % error margin, which is a standard and acceptable benchmark in heat transfer predictions to assess the prediction accuracy (Qiu et al., 2024; Mudawar et al., 2024b; Zhang et al., 2015; Li et al., 2018). The key performance metrics, namely, MAE, RMSE, and $R^2$, are displayed for the quantitative evaluation of the model's accuracy. Fig. 5 (a) shows the performance of BR-BPNN on the subcritical training dataset. The $R^2$ value of 0.9908 indicates a strong correlation between actual and predicted values, with an MAE of 3.23 °C (0.83 %) and RMSE of 12.36 °C (3.20 %). The majority of data points cluster tightly along the diagonal line, with most predictions residing within the ±10 % margin. The phase change phenomena in the vertical

tubes led to the dryout phenomenon and an abrupt increase in the wall temperature at that location for subcritical flows. This highly non-linear behavior of the wall temperature was effectively captured by the proposed BR-BPNN model.

The BR-BPNN's generalization capability was further assessed using isolated unseen subcritical testing data, as shown in Fig. 5(b). $R^2$ remained strong at 0.9504, but its value was reduced compared to the training data because the testing data were completely unseen. The MAE was 12.44 °C (2.84 %), and RMSE was 28.75 °C (6.57 %). Despite this, 96.19 % of testing data points remained within the ±10 % error margin and 98.83 % of the data remained within the ±20 % error margin, which demonstrates reasonable prediction accuracy. Slight deviations outside the ±10 % error band can be attributed to the inherent complexity of subcritical boiling phenomena, coupled with experimental uncertainties among different sources and the model's limitations in generalizing across regions with sparse or noisy data.

Fig. 5(c) shows the performance of BR-BPNN on the supercritical training data. The model showed excellent accuracy in terms of performance metrics, with an $R^2$ of 0.9924, MAE of 2.27 °C (0.56 %), and RMSE of 3.66 °C (0.91 %). Almost all the data points remained within the ±10 % error margin, clustering tightly around the diagonal line. The superior performance of the model is due to the inherently smoother and continuous nature of the heat transfer mechanism observed in supercritical flow behavior.

The scatter plot for the supercritical testing data, as shown in Fig. 5 (d), demonstrates the ability of the BR-BPNN to generalize to unseen supercritical data. The $R^2$ of 0.9401, MAE of 13.37 °C (2.99 %) and RMSE of 20.17 °C (4.52 %) are slightly lower compared to those of the training data, as expected, but overall values remain within acceptable range for heat transfer studies, especially related to wall temperature predictions. Overall, 97.3 % of the testing data are within the ±10 % error margin and 100 % are within the ±15 % error margin. The heat transfer characteristics in supercritical flows are fundamentally different from those in subcritical flows because there is no distinct transition from liquid to vapor, and it acts as a single fluid. The single-phase nature of fluid simplifies the learning process of neural network models, particularly for the Normal Heat Transfer (NHT) and Heat Transfer Enhancement (HTE) regimes. Although the present model has some difficulties in accurately capturing the Heat Transfer Deterioration (HTD) phenomenon, particularly near the pseudo-critical temperature range caused by rapid changes in the thermophysical properties of the fluid, the overall values of predicted wall temperature fall within the acceptable error range. These scatter plots validate the efficacy of the developed BR-BPNN model in predicting the wall temperatures for a wider range of input conditions.

### 3.2. Performance evaluation of the developed BR-BPNN model for testing data

This section describes the performance of the developed BR-BPNN model on the testing dataset. The performance evaluation was conducted using wall temperature vs local bulk enthalpy plots by comparing the experimental values with predicted ones. The testing dataset was not part of the training and was kept completely isolated (unseen) to avoid any bias and ensure the generalizability and validation of the developed model.

#### 3.2.1. Subcritical testing data

Fig. 6 illustrates a comparative analysis of the actual and BR-BPNN-predicted wall temperatures ($T_w$) as a function of bulk enthalpy for subcritical water flows, based on experimental data from Bennett et al. (1967). All tested data points in these plots were completely unseen, and none of the data points were part of the training from the entire dataset of their experiments. The four subplots in Fig. 6 depict the performance of the BR-BPNN under varying flow and heat transfer conditions, which show its predictive accuracy across a range of operating parameters. As

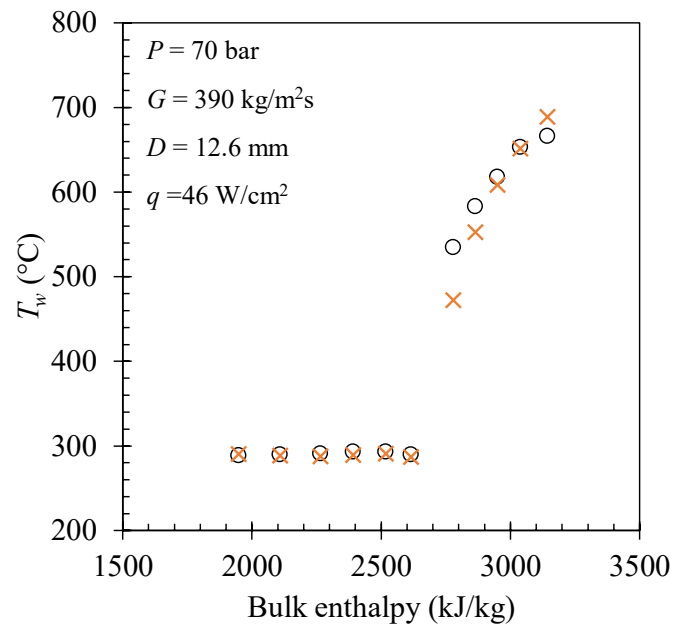


**Fig. 7.** Comparison of actual vs BR-BPNN-predicted wall temperature over given bulk enthalpy for subcritical water based on experimental data of (Bennett et al., 1967).

shown in Fig. 6, the BR-BPNN model accurately captured the steep increase in wall temperature at the dryout location. This remarkable agreement, particularly at the sharp jump, indicates the excellent ability of the model to predict the highly non-linear behavior of the wall temperature associated with the dryout phenomenon. Fig. 6(b) presents a similar trend, where the BR-BPNN demonstrates strong predictive capabilities for predicting a sharp rise in the wall temperature when the mass flow rate is reduced from 1356 to 1004 kg/m$^2$s. As shown in Fig. 6 (c), the BR-BPNN continued to perform robustly when both the mass flow rate and heat flux were simultaneously reduced to 664 kg/m$^2$s and 64.8 W/cm$^2$, respectively. The predictions of wall temperatures remain in close agreement with the experimental data, capturing their linear behavior at lower enthalpy values as well as their non-linear behavior at higher enthalpy values when approaching dryout. Fig. 6(d) illustrates the good predictive capability of the BR-BPNN while maintaining a mass flow rate of 665 kg/m$^2$s and significantly increasing the heat flux from 64.8 to 82 kW/m$^2$. The wall temperature exhibits a sharp rise up to a large temperature of approximately 780 °C.

A simultaneous reduction in both the mass flow rate and heat flux can also result in a similar wall temperature vs bulk enthalpy trend, but the post-dryout temperature continually increases, as shown in Fig. 7. The proposed BR-BPNN model can also accurately capture this trend.

A comparison between the actual and BR-BPNN-predicted wall temperatures ($T_w$) as a function of bulk enthalpy for subcritical water flows is shown in Fig. 8. The predicted wall temperatures are compared with experimental data obtained by Wang et al. (2011) in Fig. 8(a) and with those of Shen et al. (2016) conducted under relatively high-pressure conditions in Fig. 8(b–c). Fig. 8(a) shows the wall temperature variations observed in a vertical tube for the dryout condition. It can be observed that the wall temperature remains almost constant in the low-enthalpy regions and then suddenly rises sharply at the dryout point owing to the exhaustion of the liquid film at the heated wall, resulting in poor heat transfer from the wall to the vapor film. Eventually, the liquid or liquid droplets present in the core of the tube rewet the surface as they move towards the tube wall. This rewetting causes a sharp decrease in temperature above the bulk enthalpy of 2000 kJ/kg, as shown in Fig. 8 (a). The wall temperature once again starts increasing gradually for enthalpies above 2250 kJ/kg, when the liquid droplets or

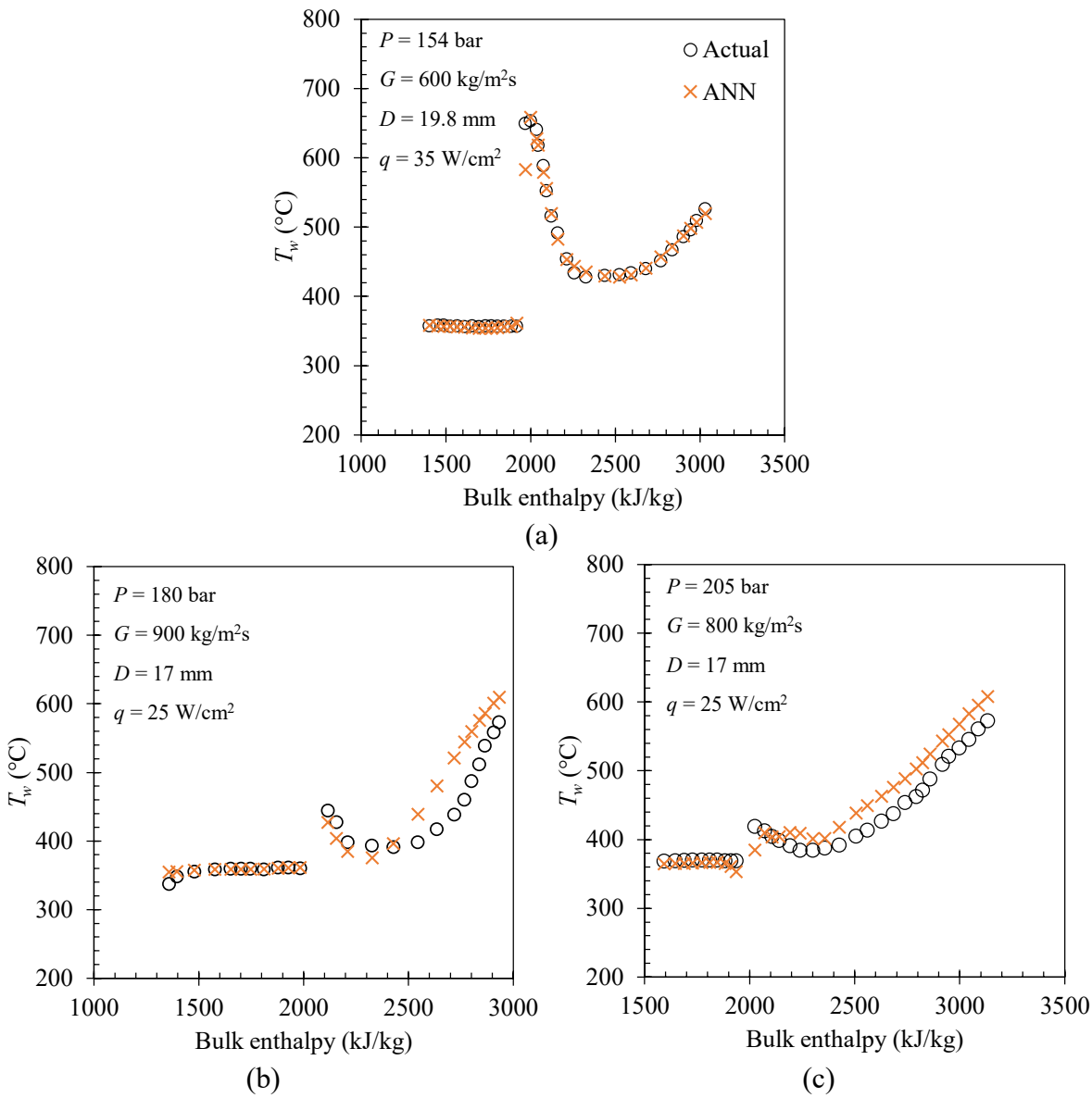


**Fig. 8.** Comparison of actual vs BR-BPNN-predicted wall temperature over given bulk enthalpy for subcritical water based on experimental data of (a) (Wang et al., 2011) and (b–c) (Shen et al., 2016).

intermittent film on the heated tube wall starts depleting owing to evaporation, and the vapor becomes more dominant again near the heated wall. This drastic variation in wall temperature was accurately captured by the developed BR-BPNN model. Such a dramatic temperature change is predominantly observed at high pressures, low heat fluxes, and low mass flow rates. To further assess the prediction accuracy, the model was tested using the experimental data of Shen et al. (2016) at pressures of 180 and 205 bar, as shown in Fig. 8(b–c). Although the model predicts the wall temperature relatively well throughout the enthalpy range considered, there was a slight deviation in the wall temperature prediction when the wall temperature increased gradually after the rewetting liquid started evaporating again, as shown in Fig. 8(b–c).

In the case of high pressure, high heat flux, and high mass flow rate, the wall temperature suddenly increases sharply at the dryout location and gradually decreases as the core liquid or liquid droplets rewet the dried-out surface. Because the flow rate is high, a sufficient liquid mass is present in the core to completely rewet the dried-out surface until the outlet, as shown in Fig. 9(a–b). The wall temperature jump observed at the dryout location for the combination of input parameters in Fig. 9(a) is higher than that in Fig. 9(b). Both these temperature profiles were remarkably captured, as shown in Fig. 9(a–b), by comparing the BR-BPNN-predicted results with the experimental results of Becker et al. (1983) and Herkenrath et al. (1967). Although the data range shown in Fig. 9 was isolated and unseen during training, the exceptional agreement between the model and actual data is because a large number of datasets available from these experiments (Becker et al., 1983; Herkenrath et al., 1967) were utilized for training.

### 3.2.2. *Supercritical testing data*

Fig. 10 shows a comparison of the actual and BR-BPNN-predicted wall temperatures ($T_w$) as a function of bulk enthalpy for supercritical water for the unseen testing data. As shown in Fig. 10, the model's performance in capturing the NHT behavior under supercritical conditions is excellent. Fig. 10(a–b) shows an experimental database collected from (Xu, 2004), which was completely unseen t in this study and was not considered for training purposes. In Fig. 10(a), the comparison is based on data at a pressure of 230 bar, mass flux of 1200 kg/m$^2$s, and

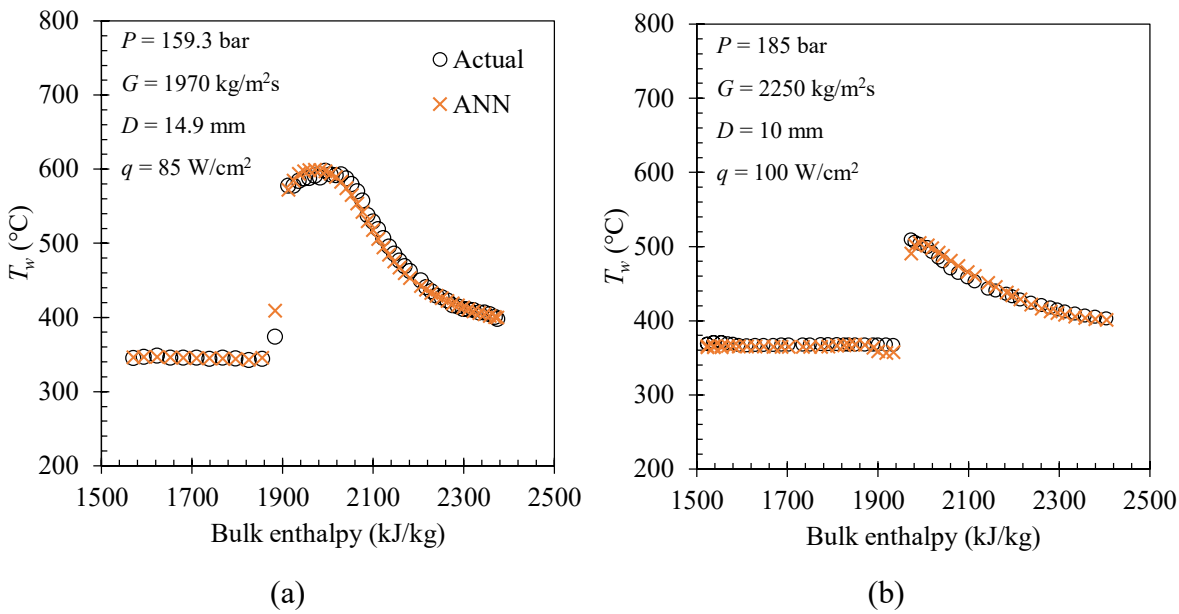


**Fig. 9.** Comparison of actual vs BR-BPNN-predicted wall temperature over given bulk enthalpy for subcritical water based on experimental data of (a) Becker et al. (Becker et al., 1983) and (b) Herkenrath et al. (Herkenrath et al., 1967).

heat flux of 60 W/cm$^2$. The wall temperature can be seen to steadily increase with bulk enthalpy, which is commonly referred to as the NHT regime. The BR-BPNN remarkably predicts the wall temperature for the entire range of bulk enthalpies across all diverse ranges of input conditions, as shown in Fig. 10. This validates the ability of the BR-BPNN to effectively model the uniform heat transfer properties observed at supercritical pressures, where phase transitions are absent and thermal properties vary smoothly. Fig. 10(b) depicts a similar comparison for conditions at a higher pressure of 250 bar but with the same mass and heat fluxes. The trend in $T_w$ is consistent with that of subplot 10(a), showing a gradual increase with bulk enthalpy according to the NHT phenomena. The BR-BPNN accurately predicted the temperature profile, closely matching the experimental data across the entire range of bulk enthalpy. Further validation of the BR-BPNN model for a normal heat transfer regime was conducted using isolated testing data obtained from Loewenberg et al. (2008), which again showed remarkably close validation.

Fig. 11 presents a comparative analysis of the experimentally determined and BR-BPNN-predicted wall temperatures ($T_w$) as a function of bulk enthalpy for supercritical water to illustrate the phenomenon of HTD. Fig. 11(a–b) is derived from the experimental datasets of Wang et al. (2011), revealing the capacity of the BR-BPNN model to replicate temperature profiles within HTD regimes across different operational parameters. As shown in Fig. 11(a), the initial jump in $T_w$ exhibits a sharp increase with increasing bulk enthalpy, reaching a maximum before a subsequent rapid decline. This behavior is caused by the transition of fluid properties as the bulk fluid temperature approaches and surpasses the pseudo-critical temperature. The sharp decrease in wall temperature after the peak indicates a reduction in thermal resistance, probably owing to the re-establishment of turbulent mixing once the fluid properties stabilize beyond the pseudo-critical point. The BR-BPNN predicted $T_w$ values accurately to capture the overall trends, including the steep rise and sharp decline. Minor deviations were observed at a few points after the peak, which can be attributed to the complexity of the rapidly changing fluid properties and flow behavior in the pseudo-critical regime. These deviations are within the acceptable error range, acknowledging the complexities involved in modeling this sensitive heat transfer area.

Fig. 11(b) shows an initial gradual increase in $T_w$ with increasing bulk enthalpy, followed by a sudden rapid increase in the HTD region. Beyond HTD, the temperature increased again and gradually increases with a smaller gradient. This indicates a persistent reduction in heat transfer efficiency, suppression of turbulent mixing, and laminarization of the flow adjacent to the heated surface. The BR-BPNN model accurately captures the temperature profile up to the HTD region but slightly deviates when HTD occurs. This indicates that the model has limitations in accurately capturing the HTD region in this specific case, primarily due to the limited number of training data points available in that regime. However, this issue can be effectively addressed by incorporating high-resolution data into the HTD region across a broader range of input conditions. Fig. 11(c) shows that the BR-BPNN struggles to predict the wall temperature accurately for a few data points immediately before a sharp temperature increase is observed. Overall, the BR-BPNN model can predict the wall temperature extremely well across the investigated ranges of bulk enthalpy, effectively capturing trends, including rapid temperature transitions of rise and fall. However, it also highlights the scope for further improvement, particularly before or after the post-HTD regime, wherein the rapid change in fluid properties and flow instabilities add complexities to accurate modeling.

## 4. Conclusions

This paper presented the development and application of a BR-BPNN model to predict the wall temperatures in vertical tubes under subcritical and supercritical water flow for given input parameters of pressure, mass flow rate, tube diameter, heat flux, and bulk enthalpy. The following key conclusions were drawn.

- The developed BR-BPNN model showed promising accuracy in the prediction of wall temperature for a diverse range of input conditions under both subcritical and supercritical flow conditions. To assess the performance of the developed model, a k-fold cross-validation method was applied before training on the full dataset. The hyperparameter tuning study revealed that the optimal neural network architecture consisted of four layers with 125, 100, 75, and 50 neurons in each layer. Further expansion of the network architecture by adding more layers and neurons resulted in performance deterioration while significantly prolonging the training process.
- A total of 34 477 data points from subcritical and 5535 data points from supercritical conditions were utilized from the experimental

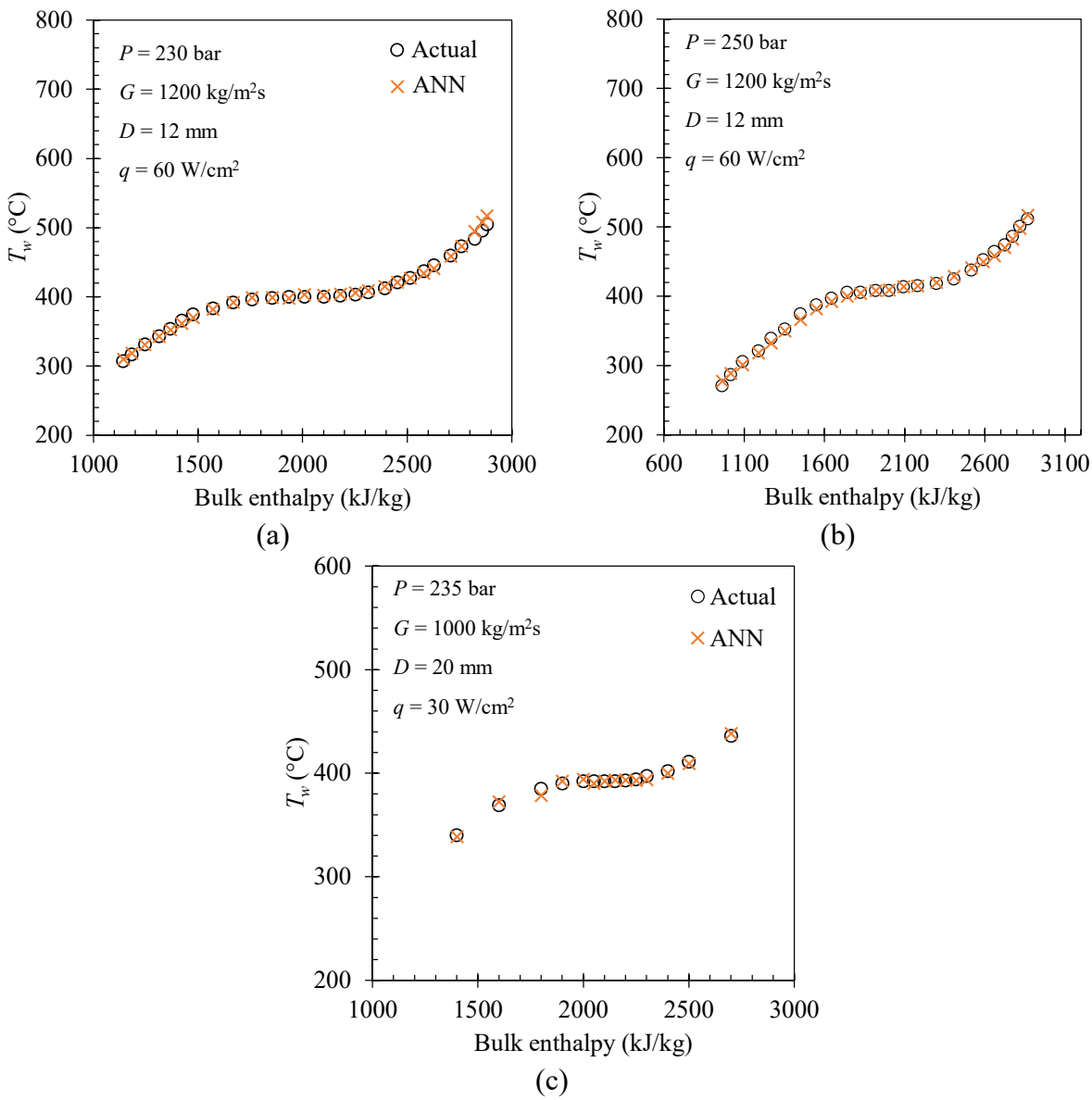


**Fig. 10.** Comparison of actual vs BR-BPNN-predicted wall temperature over given bulk enthalpy, showing NHT for supercritical water based on experimental data of (a–b) Xu (Xu, 2004) and (c) Loewenberg et al. (Loewenberg et al., 2008).

database to train the model. The model achieved an $R^2$ of 0.9504, RMSE of 6.57 %, and MAE of 2.84 % for unseen testing data of subcritical flow, with 96.19 % of the data remaining within the ±10 % error margin. Meanwhile, the model achieved an $R^2$ of 0.9401, RMSE of 4.52 %, and MAE of 2.99 % for unseen testing data of supercritical flow, with 97.3 % of the data remaining within the ±10 % error margin.

- The developed model effectively captures highly non-linear behaviors, such as the steep temperature rise observed at the dryout location and the subsequent fall in temperature in the fully developed region owing to rewetting of the heated tube surface, as commonly seen in subcritical flows. The model also predicted the rerise in wall temperature once the rewetting film began exhausting on the tube wall. The present model can also effectively capture the Noraml Heat Transfer (NHT) and Heat Transfer Deterioration (HTD) phenomena commonly observed in supercritical flows.

The unified ANN model in this study provides a quick and precise method for estimating wall temperatures in both the subcritical and supercritical flow regimes. For design engineers, this model significantly accelerates safety analysis and design optimization processes by enabling rapid parametric evaluations, such as changes in tube diameter, heat flux, mass flow rate, and pressure, which would otherwise be computationally costly using traditional CFD techniques. Moreover, the presented ANN model can be incorporated as a closure correlation in thermal-hydraulics codes like Reactor Excursion and Leak Analysis (RELAP) and TRAC/REPAL Advanced Computational Engine (TRACE). The model can be utilized in real-time predictive and monitoring systems to ensure the operational safety of critical industrial components, such as boilers, heat exchangers, and nuclear reactors. An example of real-world use is the implementation of feedback control loops for coolant flow rates, where the ANN's real-time predictions could foresee and stop dangerous situations, such as sudden spikes in wall temperature caused by insufficient flow or excessive heating.

Despite the excellent performance of the proposed BR-BPNN model, certain limitations were encountered. Specifically, the model showed reduced accuracy in the HTD region under supercritical flow conditions, primarily due to the limited density of high-resolution experimental

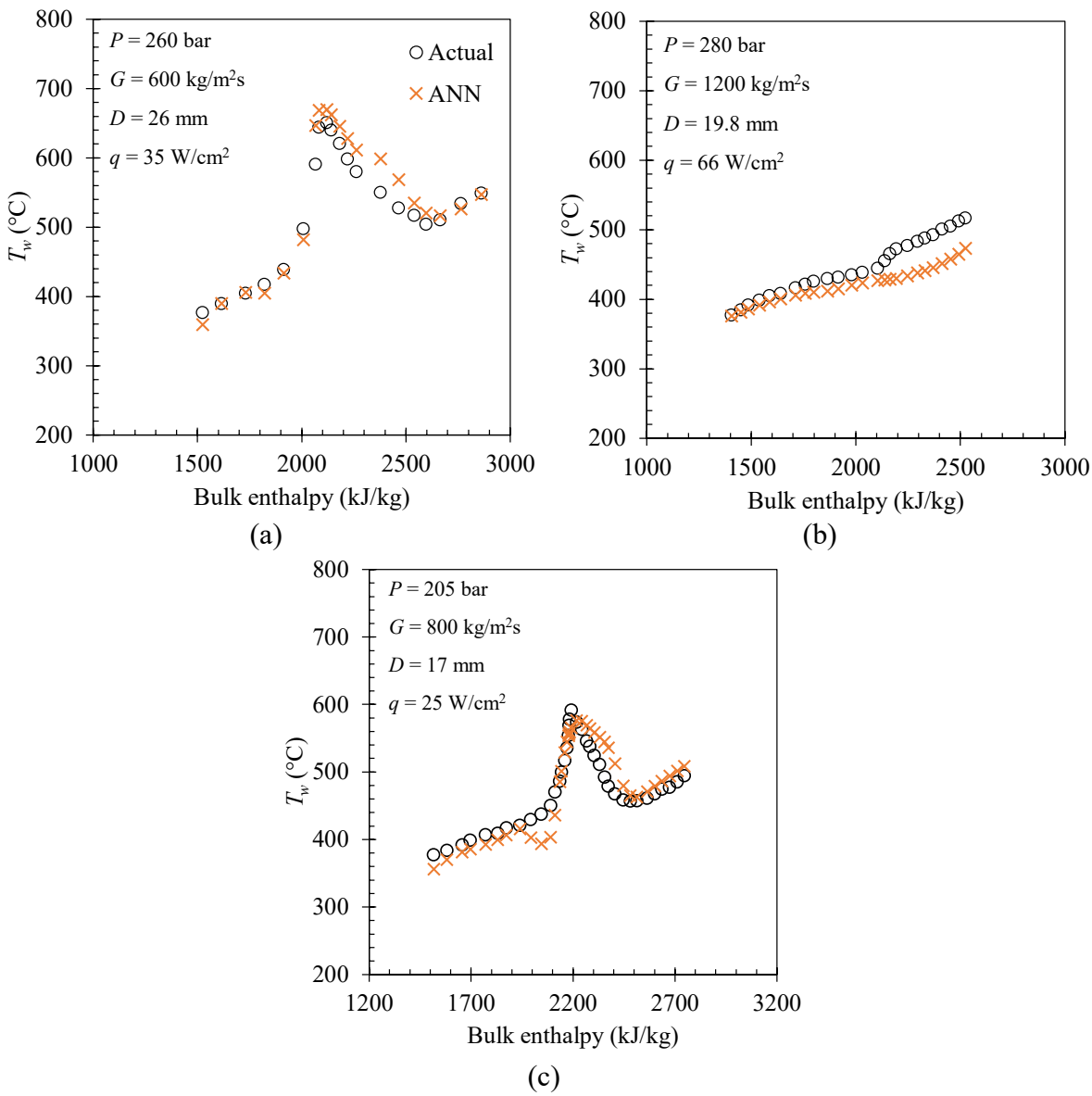


**Fig. 11.** Comparison of actual vs BR-BPNN-predicted wall temperature over given bulk enthalpy, showing HTD for supercritical water based on experimental data of (a–b) Wang et al. (Wang et al., 2011) and (c) Lei et al. (Lei et al., 2017).

data in this narrow and highly non-linear regime. A training dataset obtained either from experimental or CFD studies, particularly around the pseudo-boiling region, can be added to the training dataset as part of future work so that the model can learn the HTD pattern well. Moreover, the same methodology can be extended to more complex geometries like annular channels and rod bundles by adding further input parameters like hydraulic diameter, rod pitch, or channel gap size. Transient variations of wall temperature under different operating conditions are also promising directions for future studies that can be directly applied to real-time monitoring, diagnostics, and digital twin applications.

## CRediT authorship contribution statement

**Liril D. Silvi:** Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **A.R. Shanmugam:** Writing – review & editing, Visualization. **Ki Sun Park:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgment

This research was supported by the United Arab Emirates University under Grant Nos. G00004258 and G00004882.

## Data availability

Data will be made available on request.

## References

Aliyu, A.M., Choudhury, R., Sohani, B., Atanbori, J., Ribeiro, J.X.F., Ahmed, S.K.B., Mishra, R., 2023. An artificial neural network model for the prediction of entrained droplet fraction in annular gas-liquid two-phase flow in vertical pipes. Int. J.

Multiphas. Flow 164, 104452. https://doi.org/10.1016/J.IJMULTIPHASEFLOW.2023.104452.
Beale, M.H., Hagan, M.T., Demuth, H.B., 2020. Deep learning toolbox. R2020a User's Guide. The MathWorks, Inc., Natick, MA, USA.
Becker, K.M., Ling, C.H., Hedberg, S., Strand, G., 1983. An experimental investigation of post dryout heat transfer. https://inis.iaea.org/collection/NCLCollectionStore/_Public/15/049/15049472.pdf. (Accessed 24 December 2024).
Bennett, A.W., Hewitt, G.F., Kearsey, H.A., Keeys, R.K.F., 1967. Heat transfer to steam-water mixtures flowing in uniformly heated tubes in which the critical heat flux has been exceeded. In: Proceedings of the Institution of Mechanical Engineers, Conference Proceedings. SAGE Publications, London, England, pp. 258–267.
Burden, F., Winkler, D., 2008. Bayesian regularization of neural networks. Methods Mol. Biol. 458, 23–42. https://doi.org/10.1007/978-1-60327-101-1_3.
Chang, W., Chu, X., Binte Shaik Fareed, A.F., Pandey, S., Luo, J., Weigand, B., Laurien, E., 2018. Heat transfer prediction of supercritical water with artificial neural networks. Appl. Therm. Eng. 131, 815–824. https://doi.org/10.1016/J.APPLTHERMALENG.2017.12.063.
Demuth, H.B., Hagan, M.T., Beale, M.H., De Jesús, O., 2014. In: Hagan, Martin (Ed.), Neural Network Design. Stillwater, OK, United States.
Géron, A., 2019. Hands-on Machine Learning with Scikit-Learn, Keras, and Tensorflow. O'Reilly Media. https://www.oreilly.com/library/view/hands-on-machine-learning/9781492032632/. (Accessed 14 January 2025).
Gouravaraju, S., Narayan, J., Sauer, R.A., Gautam, S.S., 2023. A Bayesian regularization-backpropagation neural network model for peeling computations. J. Adhes. 99, 92–115. https://doi.org/10.1080/00218464.2021.2001335.
Herkenrath, H., Moerk-Moerkenstein, P., Jung, U., Weckermann, F.-, 1967. Wärmeübergang an wasser bei erzwungener strömung im druckbereich von 140 bis 250 bar. darstellung der versuchsergebnisse. (Heat Transfer in Water With Forced Circulation in the 140 to 250 Bar Pressure Range. Results of Investigations). European Atomic Energy Community, Ispra (Italy).
Kim, H., Moon, J., Hong, D., Cha, E., Yun, B., 2021. Prediction of critical heat flux for narrow rectangular channels in a steady state condition using machine learning. Nucl. Eng. Technol. 53, 1796–1809. https://doi.org/10.1016/J.NET.2020.12.007.
Lei, X., Li, H., Zhang, W., Dinh, N.T., Guo, Y., Yu, S., 2017. Experimental study on the difference of heat transfer characteristics between vertical and horizontal flows of supercritical pressure water. Appl. Therm. Eng. 113, 609–620. https://doi.org/10.1016/J.APPLTHERMALENG.2016.11.051.
Li, J., Zhou, T., Ju, Z., Huo, Q., Xiao, Z., 2014. Sensitivity analysis of CHF parameters under flow instability by using a neural network method. Ann. Nucl. Energy 71, 211–216. https://doi.org/10.1016/J.ANUCENE.2014.03.040.
Li, Q., Avramova, M., Jiao, Y., Chen, P., Yu, J., Pu, Z., Chen, J., 2018. CFD prediction of critical heat flux in vertical heated tubes with uniform and non-uniform heat flux. Nucl. Eng. Des. 326, 403–412. https://doi.org/10.1016/J.NUCENGDES.2017.11.009.
Liang, X., Xie, Y., Day, R., Meng, X., Wu, H., 2021. A data driven deep neural network model for predicting boiling heat transfer in helical coils under high gravity. Int J Heat Mass Transf 166, 120743. https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2020.120743.
Lin, L., Gao, L., Kedzierski, M.A., Hwang, Y., 2022. A general model for flow boiling heat transfer in microfin tubes based on a new neural network architecture. Energy and AI 8, 100151. https://doi.org/10.1016/J.EGYAI.2022.100151.
Loewenberg, M.F., Laurien, E., Class, A., Schulenberg, T., 2008. Supercritical water heat transfer in vertical tubes: a look-up table. Prog. Nucl. Energy 50, 532–538. https://doi.org/10.1016/J.PNUCENE.2007.11.037.
MacKay, D.J.C., 1992. A practical bayesian framework for backpropagation networks. In: Neural Comput. MIT Press, pp. 448–472. https://doi.org/10.1162/NECO.1992.4.3.448.
Mudawar, I., Darges, S.J., Devahdhanush, V.S., 2024a. Prediction technique for flow boiling heat transfer and critical heat flux in both microgravity and Earth gravity via artificial neural networks (ANNs). Int J Heat Mass Transf 220, 124998. https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2023.124998.
Mudawar, I., Darges, S.J., Devahdhanush, V.S., 2024b. Prediction technique for flow boiling heat transfer and critical heat flux in both microgravity and Earth gravity via artificial neural networks (ANNs). Int J Heat Mass Transf 220, 124998. https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2023.124998.
Niu, S., Bi, J., Li, Y., Lu, G., 2024. Prediction of critical heat flux and position in narrow rectangular channels using deep feed-forward neural networks coupling with empirical correlations. Int J Heat Mass Transf 221, 125042. https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2023.125042.
Park, H.M., Lee, J.H., Kim, K.D., 2020. Wall temperature prediction at critical heat flux using a machine learning model. Ann. Nucl. Energy 141, 107334. https://doi.org/10.1016/J.ANUCENE.2020.107334.
Qiu, Z., Ma, Y., Huang, T., Deng, J., Kong, D., Wu, D., Zhang, J., 2024. Development and application of data-driven CHF model in system analysis code. Nucl. Eng. Des. 428, 113488. https://doi.org/10.1016/J.NUCENGDES.2024.113488.
Sajjad, U., Yan, W.M., Hussain, I., Mehdi, S., Sultan, M., Ali, H.M., Said, Z., Wang, C.C., 2024. Physics and correlations informed deep learning to foresee various regimes of the pool boiling curve. Eng. Appl. Artif. Intell. 136, 108867. https://doi.org/10.1016/J.ENGAPPAI.2024.108867.
Shen, Z., Yang, D., Chen, G., Xiao, F., 2014. Experimental investigation on heat transfer characteristics of smooth tube with downward flow. Int J Heat Mass Transf 68, 669–676. https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2013.09.045.
Shen, Z., Yang, D., Xie, H., Nie, X., Liu, W., Wang, S., 2016. Flow and heat transfer characteristics of high-pressure water flowing in a vertical upward smooth tube at low mass flux conditions. Appl. Therm. Eng. 102, 391–401. https://doi.org/10.1016/J.APPLTHERMALENG.2016.03.150.
Varma, S., Simon, R., 2006. Bias in error estimation when using cross-validation for model selection. BMC Bioinf. 7, 1–8. https://doi.org/10.1186/1471-2105-7-91/FIGURES/4.
Wang, J., Li, H., Yu, S., Chen, T., 2011. Investigation on the characteristics and mechanisms of unusual heat transfer of supercritical pressure water in vertically-upward tubes. Int J Heat Mass Transf 54, 1950–1958. https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2011.01.008.
Wang, E., Zhao, M., Shi, H., Zhang, H., Yang, Y., Wen, Q., 2025. Critical heat flux prediction through machine learning model for narrow rectangular channels. Int. J. Therm. Sci. 216, 110050. https://doi.org/10.1016/J.IJTHERMALSCI.2025.110050.
Xu, F., 2004. Study of Water Flow and Heat Transfer Characteristics Through Pipes Under Supercritical Pressure. Xi'an Jiaotong University.
Zhang, R., Cong, T., Tian, W., Qiu, S., Su, G., 2015. Prediction of CHF in vertical heated tubes based on CFD methodology. Prog. Nucl. Energy 78, 196–200. https://doi.org/10.1016/J.PNUCENE.2014.10.001.
Zhu, X., Bi, Q., Yang, D., Chen, T., 2009. An investigation on heat transfer characteristics of different pressure steam-water in vertical upward tube. Nucl. Eng. Des. 239, 381–388. https://doi.org/10.1016/J.NUCENGDES.2008.10.026.